DE PI

DEPI | Digital Egypt Pioneers Initiative
Track: Data Engineering

Web-to-Warehouse Product Intelligence Pipeline

Compliance-first ETL | Kimball warehouse | Data-quality gate | REST API | React dashboard

35	29/23	12x6	216	550	583
ingestion sources	tables / views	DQ rules x dimensions	REST operations	catalogued features	automated tests

01 Overview
Web-to-Warehouse product intelligence

- permitted web sources
- retailer catalog

02 Problem
Manual collection is slow and inconsistent

- stale spreadsheets
- no history

03 Solution
Compliance-first ETL pipeline

- robots.txt aware
- deduplicated

04 Architecture
Airflow orchestrates 9 stages

- ingest, clean, load
- detect, reconcile

05 Warehouse
28 tables, 20 analytical views

- Alembic-managed schema
- PostgreSQL and MySQL twins

06 Data Quality
12 rules across 6 dimensions

- score persisted per run
- 98.26 measured

07 Forecasting
Prices, anomalies, advice

- damped Holt-Winters
- MAD, sigma and IQR detectors

08 Operations
173 REST operations, 23 routers

- JWT, TOTP and scoped keys
- leased jobs and SSE

09 Dashboard
20 screens, 5 themes, 4 report formats

- command palette, installable PWA
- HTML, JSON, CSV and PDF

10 Evidence
408 tests, 104 smoke checks, CI green

- mypy and Ruff clean
- 30 documents, 80 diagrams

TECH STACK

Python 3.12 · FastAPI · Airflow 2.10 · SQLAlchemy · NumPy · Alembic · WeasyPrint · React 19 · TypeScript · Tailwind · Recharts · PostgreSQL 16 · MySQL 8.4 · Argon2id + TOTP · Server-Sent Events · Docker Compose · GitHub Actions · pytest + mypy + Ruff

DEVELOPMENT TIMELINE

Plan → Design → Build → Test → Deploy → Maintain

PROJECT OUTCOMES

- Working pipeline
- 28-table warehouse, 20 views
- 12-rule DQ gate
- REST API, 173 operations
- 408 automated tests
- 104 API smoke checks
- React dashboard, 20 screens
- Alembic-managed schema
- 30 documents, 80 diagrams
- PDF and CSV reports

github.com/Ahmedbakr78/Web-to-Warehouse-Product-Intelligence-Pipeline